Distinct Topic Distributions Across Healthcare Sub-Domains in Industry 4.0: A Topic Modeling-Based Approach

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Abstract

Industry 4.0, encompassing IoT, Big Data, Blockchain, Cloud Computing, and AI, is transforming healthcare into "Healthcare 4.0," enabling personalized, data-driven care through applications like remote monitoring and advanced diagnostics. However, its integration across healthcare sub-domains presents unique challenges.

This study explores the topic distributions in Industry 4.0 healthcare literature using topic modeling to uncover trends in cardiology, oncology, orthopedics, and health records. By analyzing categorized datasets and applying Latent Dirichlet Allocation (LDA), the research maps the impact of these technologies, offering actionable insights to optimize healthcare systems and advance Healthcare 4.0 applications.

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1 Related Work

As Industry 4.0 technologies revolutionize healthcare, the integration of IoT, Big Data Analytics, Blockchain, Cloud Computing, and AI is shaping "Healthcare 4.0," enabling applications like remote patient monitoring and data-driven diagnostics for personalized care. However, understanding their application across healthcare sub-domains remains a critical challenge.

Digital technologies significantly impact operational efficiency and economic outcomes in various sectors, optimizing resource allocation and cost reduction in healthcare [1]. Studies show the transformative role of these technologies in Healthcare 4.0, with data and text mining supporting data-driven decisions to enhance patient outcomes [4]. The Covid-19 pandemic accelerated the adoption of IoT, AI, and machine learning for pandemic management [14]. Edge computing, such as the BodyEdge architecture, aids healthcare by enabling local data processing and reducing latency [15].

Text mining, including topic modeling with Latent Dirichlet Allocation (LDA), can efficiently identify key topics, providing insights for further research and application in fields like bioinformatics and social network analysis [5][13].

2 Approach

This study employs Latent Dirichlet Allocation (LDA), a probabilistic topic modeling technique, to analyze the thematic structure of Industry 4.0 applications in healthcare literature. Complementary approaches, including KeyBERT and TF-IDF, act as baselines to validate the relevance and quality of topics generated by the LDA model. The ultimate goal is to create a generalized framework for exploratory literature reviews in emerging technological fields.

2.1 Latent Dirichlet Allocation (LDA)

LDA is a three-level hierarchical Bayesian model designed for discrete data like text corpora. LDA assumes the following generative process for each document w in a corpus D:

1. Choose $N \sim \text{Poisson}(\xi)$.
2. Choose $\theta \sim \text{Dir}(\alpha)$.
3. For each of the N words w_n:
   (a) Choose a topic $z_n \sim \text{Multinomial}(\theta)$.
   (b) Choose a word w_n from $p(w_n | z_n, \beta)$, a multinomial probability conditioned on the topic z_n.

Figure 1: Generative process for LDA

This process is depicted as a probabilistic graphical model in Figure 2.

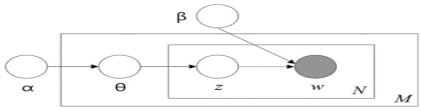

Figure 2: Graphical model for LDA

2.2 KeyBERT

The KeyBERT model was utilized to extract 70 key phrases for each document with an n-gram range of [1, 3]. This serves as a baseline to evaluate the keywords' relevance and their alignment with the topics identified by the LDA model. By comparing the output of the KeyBERT model with the LDA-generated topics, we can assess the LDA model's effectiveness in capturing key themes.

2.3 TF-IDF

The TF-IDF matrix, used as input for the LDA model, also acts as a baseline. By examining the term frequency and importance in the documents before running the LDA model, we can establish a foundational understanding of word significance within the corpus.

3 Experiment details

3.1 Dataset

The data collection process involved searching academic databases like IEEE Xplore, Springer, and Elsevier using keywords such as

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https://doi.org/10.1145/nnnnnnn.nnnnnnn

"Cardiology 4.0," "Oncology in Industry 4.0," "Orthopedics technology trends," and "Pathology advancements with Industry 4.0." Papers unrelated to these medical domains or lacking a focus on Industry 4.0 technologies were excluded, along with non-English studies. The final dataset consisted of 80 research papers, with 20 papers per subdomain (cardiology, oncology, orthopedics, and pathology), specifically selected for analyzing the impact of Industry 4.0 technologies.

3.2 Preprocessing

The dataset was further preprocessed to prepare it for input to the LDA model, which cannot directly accept raw textual input. Preprocessing included multiple steps:

(1) Text normalization
(2) Removal of punctuation and digits
(3) Stopword removal
(4) Lemmatization

3.3 Topic modelling

To further highlight important words in each document, keywords were extracted using the KeyBERT model.

(1) KeyBERT Model Configuration: The KeyBERT model was set to extract 70 key phrases per document, using an n-gram range of [1, 3].

(2) LDA model: The processed text from each document is then converted to a matrix of TF-IDF features. This matrix is used by the LDA model as an input. The total number of topics to be identified also needs to be provided as a parameter to the LDA model. The optimal number of topics for this set of documents was decided to be four after testing out different topic numbers. The model was run for 200 iterations and top 4 words were identified for each topic. The documents were then analysed based on the topics identified. The topic with maximum probability was assigned as the major theme for that document.

4 Results

4.1 Cardiology

The LDA analysis identified four key topics, with **Topic 2** (*iot, ieee, authorized, licensed*) showing the highest coherence score of 0.4326, highlighting the crucial role of IoT in healthcare. **Topic 3** (*telemedicine, ecg, cardiology, monitoring*) emphasized the growing importance of telemedicine and wearable technologies in patient monitoring.

	Topic Words	Coherence Scores
1	[artery, coronary, detection, detecting]	0.4054
2	[iot, ieee, authorized, licensed]	0.4326
3	[telemedicine, ecg, cardiology, monitoring]	0.2561
4	[solution, process, aware, integration]	0.2898

Table 1: LDA results for Cardiology

Figure 3: Word Cloud for Topic 3

In addition, KeyBERT extracted terms such as *ECG, monitoring, telemedicine, cardiology, reality*, which closely aligned with the LDA results and reinforced the focus on IoT and wearable devices. These technologies play a vital role in real-time data collection, enabling remote patient monitoring, and improving access to care, particularly within cardiology. IoT-enabled wearables, including smartwatches and health trackers, are proving essential for monitoring cardiovascular health, facilitating early diagnosis, and enhancing patient outcomes.

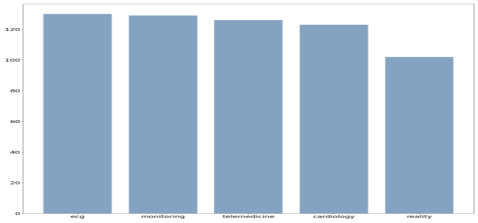

Figure 4: Top 5 keywords after KeyBERT extraction for Cardiology

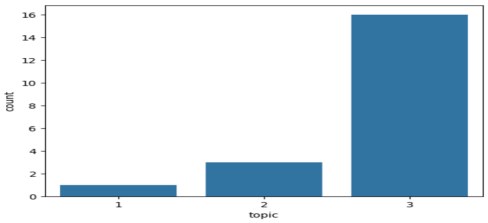

Figure 5: Distribution of documents from Cardiology corpus

Overall, the findings underscore the transformative impact of IoT and wearable technologies in shaping the future of personalized, efficient, and accessible healthcare, particularly in the cardiology domain.

4.2 Oncology

LDA identified four topics in oncology, with Topic 3 (healthcare, patient, medical, cancer) achieving the highest coherence score (0.3757), emphasizing patient-centric healthcare and medical advancements. Topic 2 (deep, cancer, image, research) highlighted the integration of AI and ML in cancer research and imaging techniques.

	Topic Words	Coherence Scores
1	[melanoma,lung,tumor,detection]	0.2988
2	[deep,cancer,image,research]	0.3112
3	[healthcare,patient,medical,cancer]	0.3757
4	[leukemia, melanoma, detection, revolution]	0.1317

Table 2: LDA results for Oncology

Figure 6: Word Cloud for Topic 3

KeyBERT extracted terms such as leukemia, cancer, detection, healthcare, melanoma, reinforcing the focus on leveraging AI and ML for early detection and diagnosis in oncology. These technologies are pivotal in analyzing medical imaging, improving diagnostic accuracy, and advancing personalized treatment strategies. AI-driven models are also being used to predict disease progression, optimize treatment protocols, and accelerate drug discovery in oncology research.

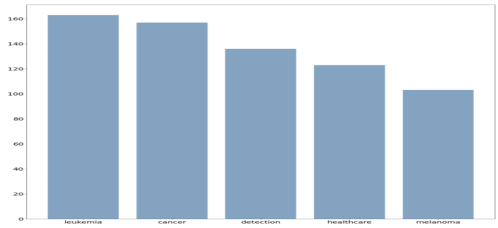

Figure 7: Top 5 keywords after KeyBERT extraction for Oncology

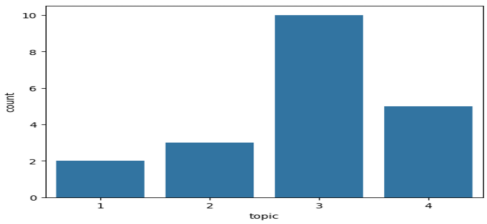

Figure 8: Distribution of documents from Oncology Corpus

The findings underscore AI and ML as transformative tools in oncology, driving innovations in cancer detection, treatment planning, and research, ultimately enhancing patient outcomes and healthcare efficiency.

4.3 Orthopedics

The orthopedics subdomain analysis uncovered pivotal trends through topic modeling. Among the four identified topics, Topic 3 (focused on surgery, robotics, and orthopedic techniques) achieved the strongest coherence score of 0.3231, reflecting the growing reliance on robotics in precision surgeries and treatments. Additionally, Topic 1 (centered on treatment, rehabilitation, and experimental approaches) and Topic 4 (rehabilitation and motion analysis) highlighted ongoing innovations aimed at enhancing recovery outcomes for patients.

	Topic Words	Coherence Scores
1	[treatment,rehabilitation,experimental,introduction]	0.2996
2	[biomechanical,leg,model,modeling]	0.2583
3	[surgery,robot,robotic,orthopedic]	0.3231
4	[rehabilitation,treatment,experimental,motion]	0.2874

Table 3: LDA results for Orthopedics

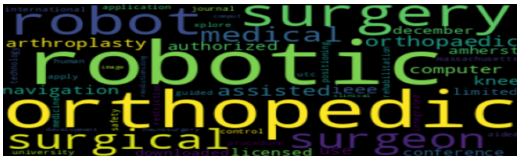

Figure 9: Word Cloud for Topic 3

Key insights derived from keyword analysis revealed terms such as "robotic surgery," "biomechanics," and "rehabilitation," illustrating how robotic systems are becoming integral to improving procedural accuracy and patient outcomes. This integration of robotics into orthopedic practices showcases a shift toward more automated, data-driven approaches in both surgical and rehabilitative care.

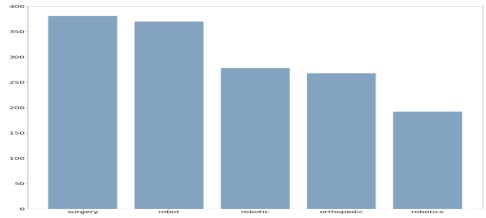

Figure 10: Top 5 keywords after KeyBERT extraction for Orthopedic

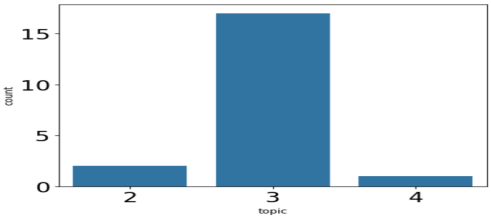

Figure 11: Distribution of documents from Orthopedic Corpus

4.4 Health Records

The exploration of health records through LDA modeling shed light on critical themes in digital health management. Topic 2, with a focus on blockchain, medical data, and healthcare systems, emphasized the increasing adoption of blockchain to address data

integrity and security concerns. While Topic 1 reflected broader applications, its coherence score of 0.3166 was indicative of its relevance across diverse implementations.

	Topic Words	Coherence Scores
1	[applying,make,involved,section]	0.3166
2	[blockchain,healthcare,medical,data]	0.2412
3	[applying,make,involved,section]	0.3166
4	[applying,make,involved,section]	0.3166

Table 4: LDA results for Health Records

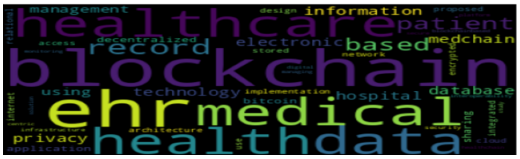

Figure 12: Word Cloud for Topic 2

The term "blockchain" emerged frequently during keyword analysis, alongside related terms such as "healthcare records" and "data privacy." These findings underline blockchain's transformative potential in health records management, offering enhanced security, traceability, and patient trust. By addressing longstanding challenges of data breaches and inefficiencies, blockchain is setting new benchmarks for the future of digital health systems.

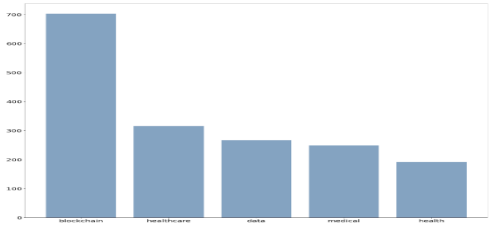

Figure 13: Top 5 keywords after KeyBERT extraction for Health Records

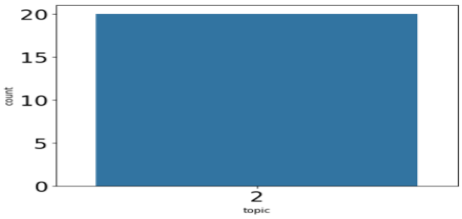

Figure 14: Distribution of documents from Health Records Corpus

5 Conclusion

This project utilized advanced topic modeling and keyword extraction to examine technological advancements in healthcare subdomains, including cardiology, oncology, orthopedics, and health records. The findings highlight the transformative impact of artificial intelligence, robotics, and blockchain in revolutionizing patient care and data management systems.

In cardiology and oncology, AI-driven diagnostics and personalized medicine emerged as pivotal innovations, improving prediction accuracy and treatment efficacy. Orthopedics exhibited significant progress in robotic-assisted surgeries and biomechanical modeling, emphasizing automation and precision in treatment and rehabilitation. Health records analysis demonstrated blockchain's capability to address critical challenges in data security, privacy, and efficiency, offering robust solutions for modernizing electronic health systems.

These results underscore the interdisciplinary integration of advanced technologies in addressing complex healthcare challenges. By enhancing precision, efficiency, and trust in healthcare systems, this work provides a comprehensive framework for understanding the potential of emerging technologies and lays a foundation for future research and innovation in healthcare.

6 References

[1]. R. Chinoracký, J. Kurotová, and P. Janošková, "Measuring the impact of digital technologies on transport industry–macroeconomic perspective," Transportation Research Procedia, vol. 55, pp. 434 441, 2021.

[2]. R. Pershina, B. Soppe, and T. Thune, "Bridging analog and digital expertise: Cross-domain collaboration and boundary-spanning tools in the creation of .

[3]. L. Barnewold and B. G. Lottermoser, "Identification of digital technologies and digitalisation trends in the mining industry," International Journal of Mining Science and Technology, vol. 30, no. 6, 2020.

[4]. T. Matsumoto, W. Sunayama, Y. Hatanaka, and K. Ogohara, "Data Analysis Support by Combining Data Mining and Text Mining," in 2017 6th IIAI International Congress on Advanced Applied Informatics (IIAI-AAI), Hamamatsu, Japan, 2017, pp. 313-318.

[5]. S. Salloum, M. Al-Emran, A. Monem, and K. Shaalan, "Using Text Mining Techniques for Extracting Information from Research Articles," 2018.

[6]. K. Lu and H. Liao, "A survey of group decision making methods in Healthcare Industry 4.0: bibliometrics, applications, and directions," Appl Intell, vol. 52, pp. 13689–13713, 2022.

[7]. Paul, M. Riffat, A. Yasir, M. N. Mahim, B. Y. Sharnali, I. T. Naheen, A. Rahman, and A. Kulkarni, "Industry 4.0 Applications for Medical/Healthcare Services," J. Sens. Actuator Netw., vol. 10, no. 3, p. 43, 2021. [Online].

[8]. A. Haleem, M. Javaid, and R. Vaishya, "Industry 4.0 and its applications in orthopaedics," J. Clin. Orthop. Trauma, vol. 10, no. 3, pp. 615–616, 2019.

[9]. X. Wei and W. Croft, "LDA-based document models for Ad-hoc retrieval," in Proceedings of the 29th Annual International ACM SIGIR, 2006.

[10]. Z. Tong and H. Zhang, "A Text Mining Research Based on LDA Topic Modelling," Computer Science and Information Technology, vol. 6, pp. 201-210, 2016. [11]. E. Ikonomakis, S. Kotsiantis, and V. Tampakas, "Text Classification Using Machine Learning Techniques," WSEAS Transactions on Computers, vol. 4, pp. 966-974, 2005.

[12]. P. Kherwa and P. Bansal, "Topic Modeling: A Comprehensive Review," ICST Transactions on Scalable Information Systems, vol. 7, p. 159623, 2018.

[13]. D. Blei, A. Ng, and M. Jordan, "Latent Dirichlet Allocation," Journal of Machine Learning Research, vol. 3, pp. 993-1022, 2003.

[14]. Moosavi, J., Bakhshi, J., and Martek, I., "The application of industry 4.0 technologies in pandemic management: Literature review and case study," Healthcare Analytics, vol. 1, p. 100008, 2021.

[15]. P. Pace, G. Aloi, R. Gravina, G. Caliciuri, G. Fortino, and A. Liotta, "An Edge-Based Architecture to Support Efficient Applications for Healthcare Industry 4.0," in IEEE Transactions on Industrial Informatics, vol. 15, no. 1, pp. 481-489, Jan. 2019.

[16] Saner, Hugo. "e-Cardiology and e-Health: from industry-driven technical progress to clinical application." European Journal of Preventive Cardiology (2014): 2-3.

[17] Ledziński, Łukasz, and Grzegorz Grześk. "Artificial intelligence technologies in cardiology." Journal of Cardiovascular Development and Disease 10.5 (2023): 202

[18] Islam, Muhammad Nazrul, et al. "Predictis: an IoT and machine learning-based system to predict risk level of cardio-vascular diseases." BMC Health Services Research 23.1 (2023): 171

[19] Bouraghi, Hamid, et al. "Virtual reality and cardiac diseases: A systematic review of applications and effects." Journal of Healthcare Engineering 2023.1 (2023).

[20] Gómez-Valiente, Patricia, et al. "Smart-IoT business process management: a case study on remote digital early cardiac arrhythmia detection and diagnosis." IEEE Internet of Things Journal(2023).

[21] Iqbal, Muhammad Javed, et al. "Clinical applications of artificial intelligence and machine learning in cancer diagnosis: looking into the future." Cancer Cell International 21.1 (2021): 270

[22] Kaul, Rohit, et al. "The role of AI for developing digital twins in healthcare: The case of cancer care." Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery 13.1 (2023).

[23] Nawreen, N., et al. "Lung Cancer Detection and Classification using CT Scan Image Processing." 2021 International Conference on Automation, Control and Mechatronics for Industry 4.0 (ACMI) (2021): 1-6

[24] Ramoliya, F., et al. "X-CaD: Explainable AI for Skin Cancer Diagnosis in Healthcare 4.0 Telesurgery." ICC 2024 - IEEE International Conference on Communications (2024): 238-243

[25] Lata, Kusum, et al. "Deep Learning-Based Brain Tumor Detection in Privacy-Preserving Smart Health Care Systems." IEEE Access (2024)

[26] Innocenti, B., Bori, E. "Robotics in orthopaedic surgery: why, what and how?" Arch Orthop Trauma Surg 141, 2035–2042 (2021)

[27] Suarez-Ahedo, C., et al. "Revolutionizing orthopedics: a comprehensive review of robot-assisted surgery, clinical outcomes, and the future of patient care." Journal of Robotic Surgery 17 (2023)

[28] Sugano, N. "Computer-assisted orthopedic surgery." J Orthop Sci 8, 442–448 (2003)

[29]Beyaz, S. "A brief history of artificial intelligence and robotic surgery in orthopedics and traumatology and future expectations." Jt Dis Relat Surg 31.3 (2020): 653-655

[30] M. Wang. "Development and Validity of Tissue Biomechanics Modeling for Virtual Robot Assisted Orthopedic Surgery System." Bioinformatics and Biomedical Engineering Conference (2009): 1-4

7 Contribution

(1) Vaishnavi Kashyap - Cardiology and Oncology implementation and results were handled
(2) Sanchita Thayagaran - Orthopedics and Health Records implementation and results were handled